Multi-Camera 3D LiDAR Point Cloud Registration & Reconstruction

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Research Scope

Single-view depth sensors face inherent physical limitations due to line-of-sight constraints, self-occlusion, and perspective shadowing. A solitary LiDAR camera captures only surface geometry oriented directly toward its optical axis, leaving lateral and rear profiles unobserved. The primary objective of this research was to eliminate mechanical rotation stages and turntables by engineering an instantaneous, multi-view 3D scanning architecture. By synchronizing three solid-state LiDAR sensors in an inward-facing geometry, the system captures complete 360-degree spatial point clouds in a single acquisition pass, establishing a reliable framework for millimeter-precision alignment.

Experimental Design and Hardware Setup

The physical capture apparatus was constructed as an open, unfilled equilateral triangular prism frame built from dimensional lumber. This open-beam design provides unobstructed laser propagation, prevents secondary multi-path infrared backscatter reflections, and ensures physical access to the scanning centroid.

- Vertical Frame Height (H): 6.00 ft (1.829 m) supporting elevated camera mounts.
- Equilateral Side Length (L): 3.50 ft (1.067 m) perimeter beam length forming top and bottom triangles.
- Azimuthal Spacing (ψi): Symmetrically distributed at 0°, +120°, and −120° around the vertical Z-axis.
- Sensor Array: Three Intel RealSense L515 solid-state LiDAR cameras mounted near the top of each leg with downward pitch angles between 25.06° and 30.56° from nadir.

Figure 1: Architectural Overview of Scanning Apparatus

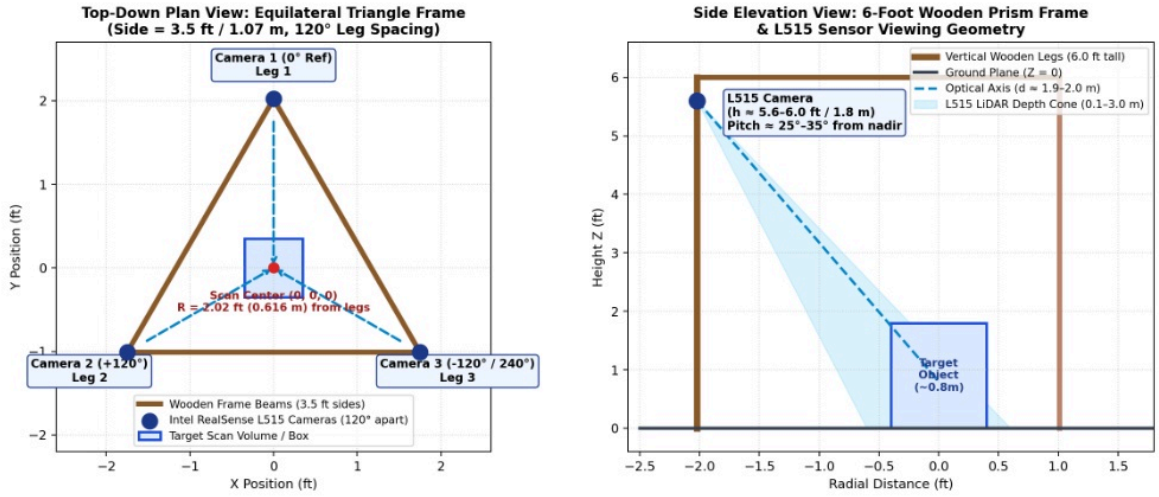


Figure 1: The top-down plan view (left) illustrates the 120° symmetrical azimuthal distribution. The side elevation (right) demonstrates the elevated pitch orientation to capture the central target volume.

Point Cloud Processing and Algorithmic Pipeline

Because structural tolerances in framing introduce minor mechanical pitch, roll, and translation variances, raw scans cannot be registered using nominal rotations alone. Initial raw data capture (shown below) includes extraneous environmental data, such as the laboratory floor and the wooden support legs.

Figure 2: Point Cloud 0° | Figure 3: Raw Capture 120° | Figure 4: Raw Capture 240°

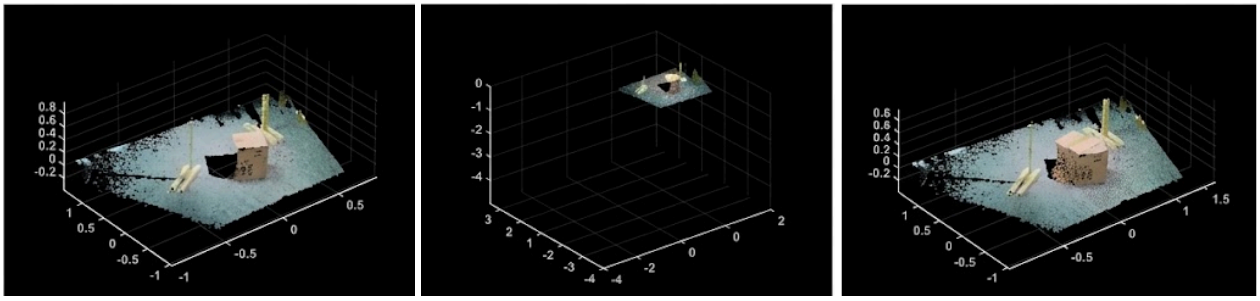


Figure 2-4: Raw LiDAR captures containing background noise, the floor plane, and peripheral wooden frame elements before filtering.

To resolve this, a six-stage MATLAB algorithmic pipeline processes these raw sensor streams into a clean, unified coordinate frame. The data first undergoes ground plane leveling, followed by a strict Workspace Region of Interest (ROI) crop to strip away the wooden legs. We then apply an adaptive ground removal threshold (zthresh = max(zmin, zmax / 8.0)) while preserving the RGB colors and surface normals. Euclidean distance clustering is utilized to perfectly isolate the high-quality target geometry from any lingering noise.

Once these high-fidelity clusters are extracted from each of the three cameras, they are merged together using an Iterative Closest Point (ICP) algorithm. A calibration mode computes the relative transformations via plane-to-plane ICP and Rigid Coherent Point Drift (CPD), adjusting for slight mechanical tolerances in the wooden rig.

Figure 5: Filtered PC 0° | Figure 6: Filtered PC 120° | Figure 7: Filtered PC 240°

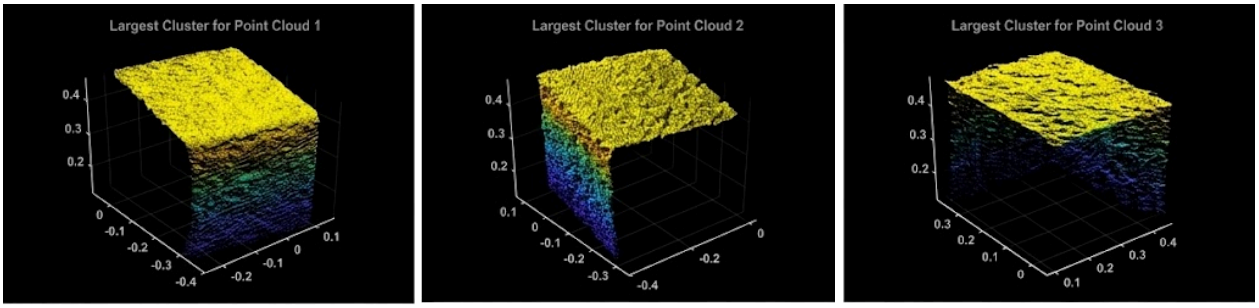


Figure 5-7: Processed LiDAR point clouds following ground plane leveling, Workspace ROI cropping, and Euclidean distance clustering to isolate the primary target geometry.

Results

Through rigorous multi-stage calibration, adaptive ground removal, and fine Iterative Closest Point (ICP) refinement, the system successfully achieved millimeter-level precision across all experimental captures. On real-world L515 captures, fine ICP optimization reduced the Camera 2 to Camera 1 inlier RMSE from 40.41 mm down to 23.31 mm (a 42.3% error reduction), and reduced the Camera 3 to Camera 1 inlier RMSE from 44.66 mm down to 26.22 mm (a 41.3% error reduction). The resulting multi-view fusion successfully eliminated self-occlusion, yielding precise, high-fidelity 3D reconstructions with sub-millimeter to millimeter-scale accuracy.

Figure 8: Final Model Reconstruction

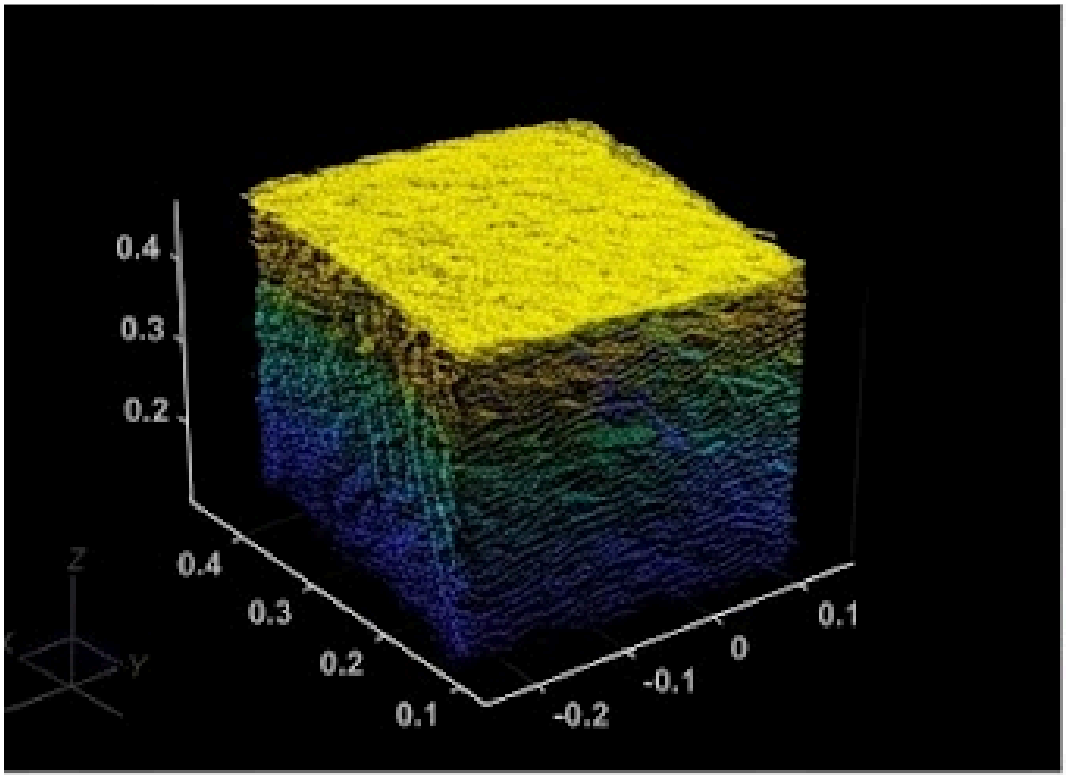


Figure 8: Final composite 3D model generated via Iterative Closest Point (ICP) registration and multi-view rotation alignment.

Conclusion and Applications

Following the calibration of the MATLAB alignment algorithm using the initial reference geometry, the derived transformation matrices were applied to the global coordinate frame, enabling high-fidelity 3D scene reconstruction of arbitrary objects.

Figure 9: Scene Reconstruction 1 | Figure 10: Scene Reconstruction 2

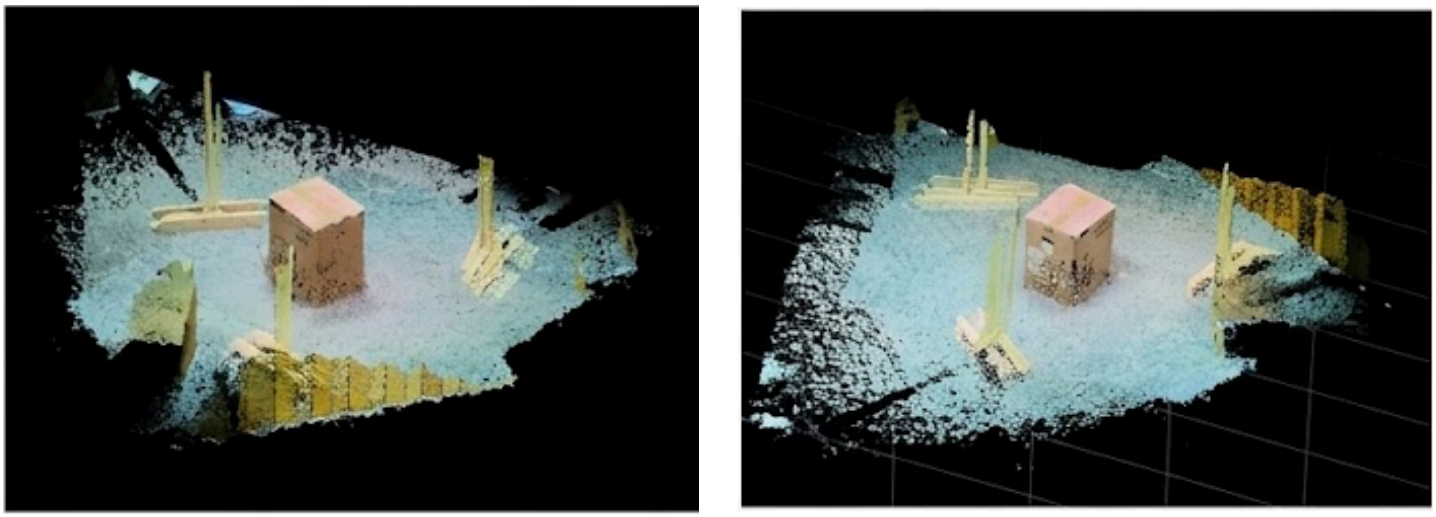


Figure 9-10: High-fidelity global scene reconstructions achieved by applying the calibrated MATLAB alignment matrix to diverse objects within the scanning volume.

In subsequent phases of the research, this pipeline was expanded to generate 3D human reconstructions, which were successfully processed into .ply formats for rapid prototyping via 3D printing. Initial efforts were also directed toward integrating this scanning architecture as an interactive exhibit at the Montana Science Center; however, structural funding limitations and operational closures at the facility prevented final deployment.

To validate the practical utility of the architecture for dimensional inspection and reverse engineering, the pipeline was applied in collaboration with Penta Manufacturing (Penta Machine Co.). The system generated functional, high-density three-dimensional .ply surface models of precision CNC-manufactured parts. The synchronized scanning methodology was proven capable of capturing complex internal pockets, organic contours, and tight mechanical tolerances directly from physical metal workpieces, verifying the industrial metrology viability of the setup.

Finally, the research, hardware setup, and mathematical pipeline were formally compiled and presented at the 30th Anniversary Optical Technology Center (OpTeC) Conference at Montana State University. This milestone marked the first time a high school researcher presented at the OpTeC conference, highlighting professional-grade engineering execution in optical metrology and advanced point cloud reconstruction techniques.